

Supporting Information for

Quantitative, Dynamic TaO_x memristor/Resistive Random Access Memory Model

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In the proposed model, the three PDEs in Eqs (2), (5), and (6) were self consistently solved through a numerical solver (COMSOL)^{S1-S4} to calculate n_D , Ψ , and T . Figure S1 shows the structure used in the model. The width (w) and depth (d) of the structure is 40 nm and 20 nm, respectively. The thicknesses of the Pd bottom electrode (BE), V_O reservoir (TaO_x) layer, switching (Ta_2O_5) layer, and the Pd top electrode (TE) layer are 35 nm, 30 nm, 5 nm, and 50 nm, respectively. The V_O density of $5 \times 10^{21} \text{ cm}^{-3}$ is chosen as a criteria for metallic CF formation^{S5}, and the maximum V_O density is $\sim 1 \times 10^{22} \text{ cm}^{-3}$. A uniform concentration of $n_D = 1 \times 10^{22} \text{ cm}^{-3}$ is assumed in the conductive TaO_x layer as an initial state. The current modulation layer (CML) height (h) is 10 nm, which is directly connected to TE.

The following assumptions are used as boundary conditions: the temperature at BE, TE and CML surfaces are 300 K, BE is always grounded during simulation, while voltage is applied to the top of CML. The definition of the parameters and constant values used in the model are summarized in Table S1 and Figure S2a. The electrical conductivity is given by the Arrhenius equation^{S6} where the pre-exponential factor σ_0 is assumed to linearly increase from 1 S/m to 10^5 S/m with increasing n_D . The electron conduction activation energy E_{AC} used in the model is 0.2 eV for the initial state where n_D is $1 \times 10^{16} \text{ cm}^{-3}$, and linearly decreases to 0.1 eV with increasing n_D up to $1 \times 10^{20} \text{ cm}^{-3}$, 0 eV with n_D up to $5 \times 10^{21} \text{ cm}^{-3}$, and -0.006 eV with n_D up to $1 \times 10^{22} \text{ cm}^{-3}$. The pre-factor of thermal conductivity k_{th0} linearly increases with increasing n_D (Figure S2b).

To introduce the compliance current (I_{CC}) during Forming and Set, the conductance of the CML is defined by Eqs (S1a) – (S1c). Here, V_I is voltage applied to the CML top surface and V_2 represents the actual voltage applied on the memristor TE, as shown in Figure S1. If the electrical field applied to CML is smaller than E_{MAX} , the conductance of CML will be I_{cc_sigma} , 10^5 S/m,

while when the electric field becomes larger than E_{MAX} the conductance of the CML is determined by Eq. S1a. During Reset, σ_{CML} is always I_{cc_sigma} , 10^5 S/m.

$$\sigma_{CML} = \frac{I_{cc} \cdot h}{|V_1 - V_2| \cdot w \cdot d}, \quad \text{if } E_{CML} \geq E_{max} \quad (S1a)$$

$$\sigma_{CML} = I_{cc_sigma}, \quad \text{if } E_{CML} < E_{max} \quad (S1b)$$

$$E_{max} = \frac{I_{cc}}{I_{cc_sigma} \cdot w \cdot d} \quad (S1c)$$

where $h = 10$ nm, $w = 40$ nm, $d = 20$ nm, $I_{CC} = 500$ μ A.

To model the transistor I - V , conductance of CML is defined in Eqs (S2a) – (S2d)^{S7}.

$$\sigma_{CML} = \left(\frac{\frac{W}{L} C_{ox} \mu \left(V_{gs} - V_t - \frac{m(V_1 - V_2)}{2} \right)}{1 + \frac{V_1 - V_2}{E_{sat} L}} \right) \frac{h}{d \cdot w \cdot (V_1 - V_2)} \quad \text{if } V_1 - V_2 < V_{dsat} \quad (S2a)$$

$$\sigma_{CML} = \left(\frac{\frac{W}{L} C_{ox} \mu \left(V_{gs} - V_t - \frac{mV_{dsat}}{2} \right)}{1 + \frac{V_{dsat}}{E_{sat} L}} + a(V_{gs} - V_t)(V_1 - V_2 - V_{dsat}) \right) \frac{h}{d \cdot w \cdot (V_1 - V_2)} \quad \text{if } V_1 - V_2 \geq V_{dsat} \quad (S2b)$$

$$V_{dsat} = \frac{2(V_{gs} - V_t)/m}{1 + \sqrt{1 + 2(V_{gs} - V_t)/mE_{sat}L}} \quad (S2c)$$

$$E_{sat} = \frac{2v_{sat}}{\mu} \quad (S2d)$$

where $L = 130$ nm, $W = 450$ nm, $m = 1.3$, $T_{ox} = 4$ nm, $\mu = 50$ cm²/(V·s), $V_T = 0.3$ V, $v_{sat} = 8 \times 10^6$ cm/s, $a = 8 \times 10^{-6}$, $h = 10$ nm, $w = 40$ nm, and $d = 20$ nm.

Figure S3 shows the I - V behavior of the CML layer. The CML layer can effectively model the effects of I_{CC} and the transistor I - V , and match well with experimental results.

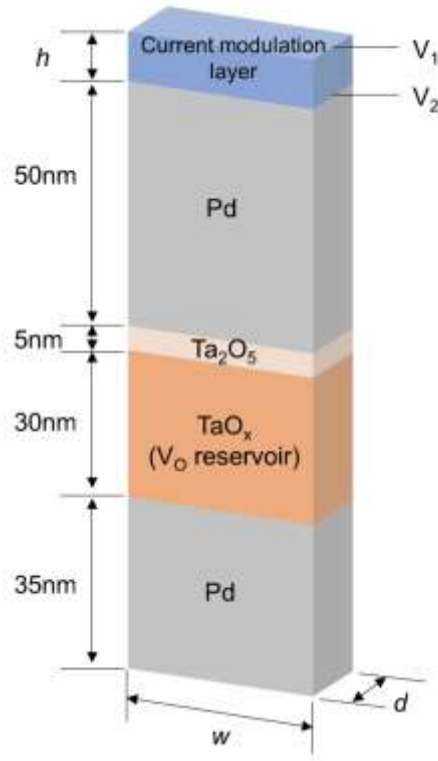


Figure S1. Device geometry used in the simulation. A uniform doping concentration of $n_D = 1 \times 10^{22} \text{ cm}^{-3}$ was assumed within the conducting TaO_x layer (V_O reservoir) at the initial state. $w = 40 \text{ nm}$, $h = 10 \text{ nm}$, $d = 20 \text{ nm}$ were used in the simulation. V_1 is the voltage applied to the current modulation layer top surface and V_2 represents the actual voltage applied to the memristor TE.

Material properties	Pd	TaO _x & Ta ₂ O ₅
Density (ρ) [kg/m ³]	11900	8200
Heat capacity (C_p) [J/(kg·K)]	50	26
Electrical conductivity (σ) [S/m]	1e7	Eq.(1)
Thermal conductivity (k) [W/(m·K)]	71.8	$k_{th0}(1 + \lambda(T - T_0))$

Constants		value
a	Hopping distance	3.2Å
f	Escape-attempt frequency	10 ¹² Hz
λ	Linear thermal coefficient	0.1
T ₀	Room temperature	293K
E _a	V _O diffusion barrier	0.85eV
α	PF coefficient1	5.48e-4
β	PF coefficient2	-5.70

Table S1. Material parameters and constants used in the proposed model.

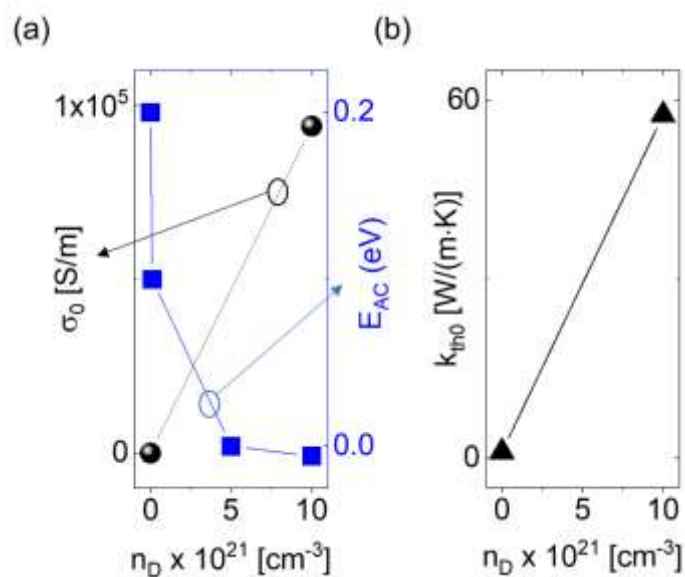


Figure S2. Parameters used in the proposed model. (a) The electrical conductivity pre-factor σ_0 and the electron conduction activation energy E_{AC} as function of n_D . (b) The thermal conductivity pre-factor k_{th0} as function of n_D .

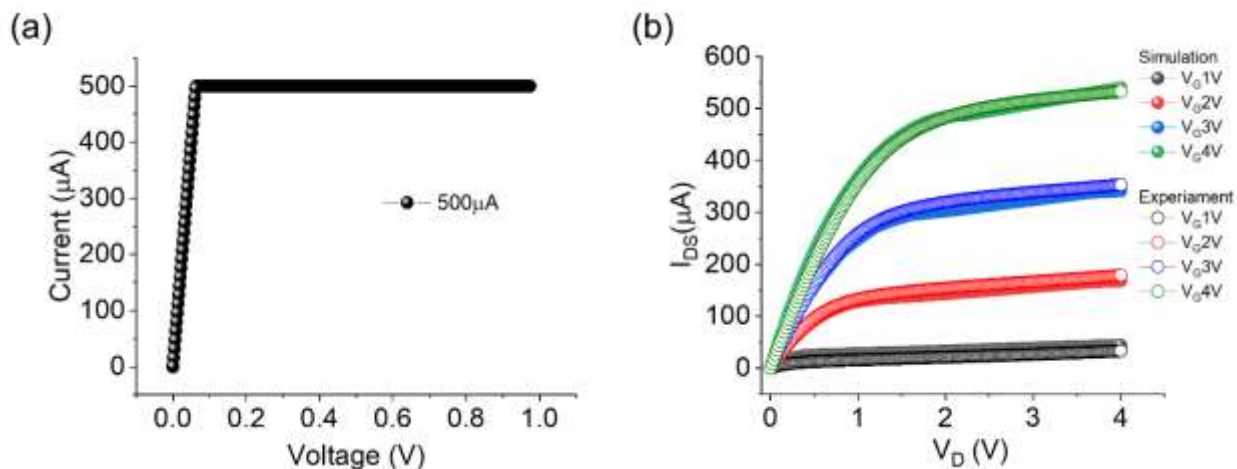


Figure S3. Current modulation layer I - V behavior, without memristor devices. (a) Simulated DC I - V with I_{CC} of 500 μA . (b) Simulated and measured transistor DC I - V curves with varying V_{G} from 1 V to 4 V. The simulation results very well matched with experimental results.

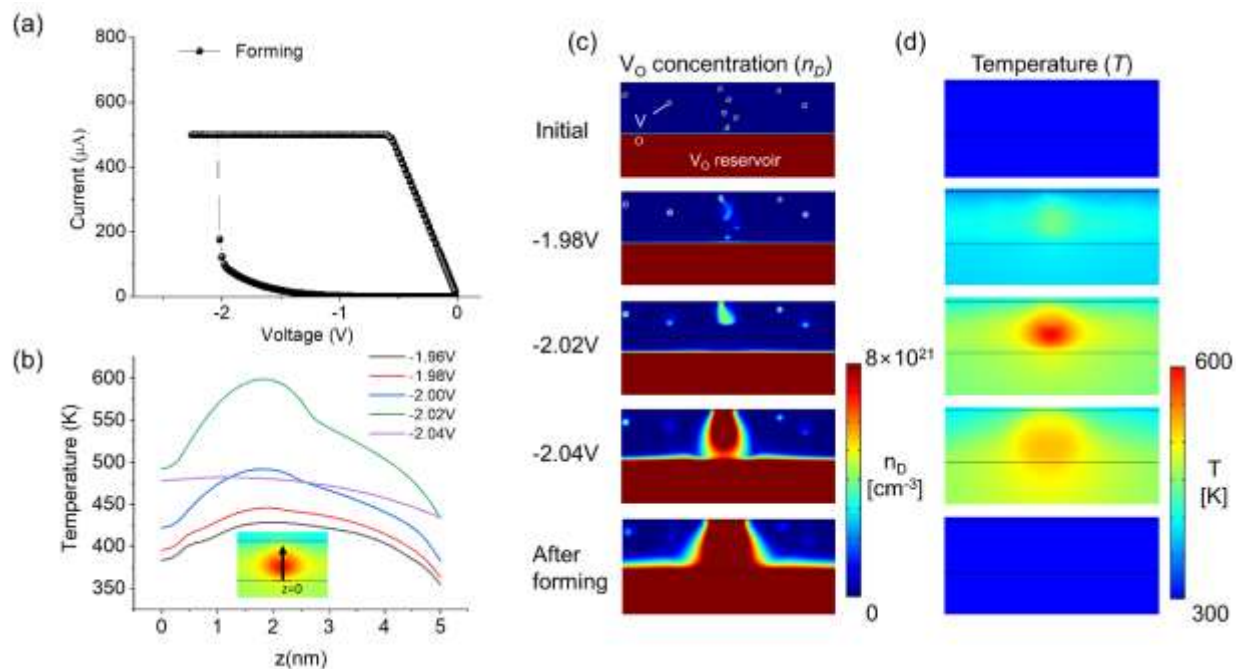


Figure S4. Evolution of V_O concentration (n_D) and temperature (T) during DC Forming. (a) DC I - V characteristics during Forming. (b) 1D profile of temperature along the z direction during Forming with different voltages. 2-D maps of (c) n_D and (d) T during Forming with different voltages.

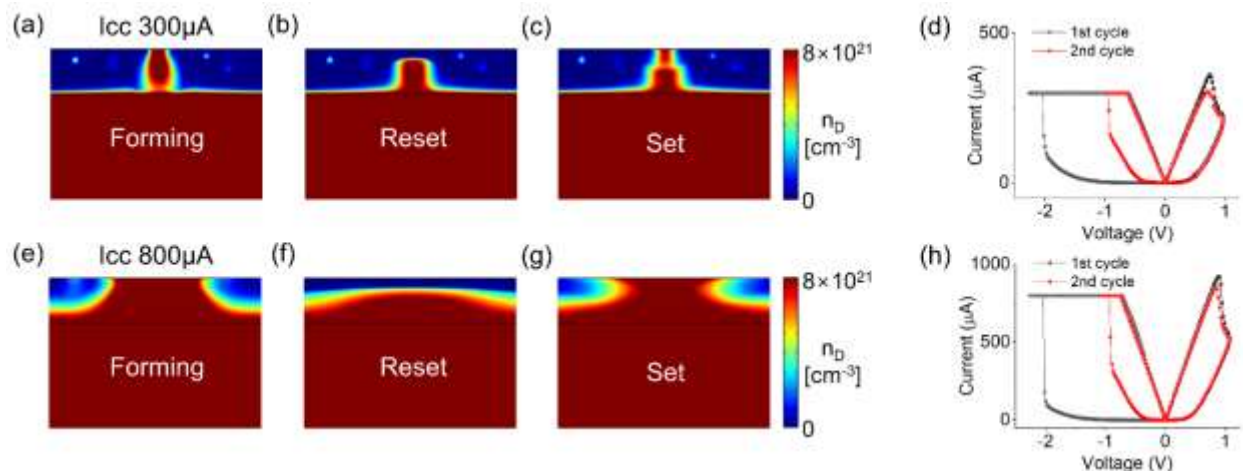


Figure S5. Simulated RS behavior with different I_{CC} s. (a-c): 2-D maps of n_D for (a) Forming, (b) Reset, and (c) Set processes with 300 μA of I_{CC} , along with the simulated DC I - V characteristics (d). (e-g) 2-D maps of n_D for (e) Forming, (f) Reset, and (g) Set process with 800 μA of I_{CC} , along with the simulated DC I - V characteristics (h).

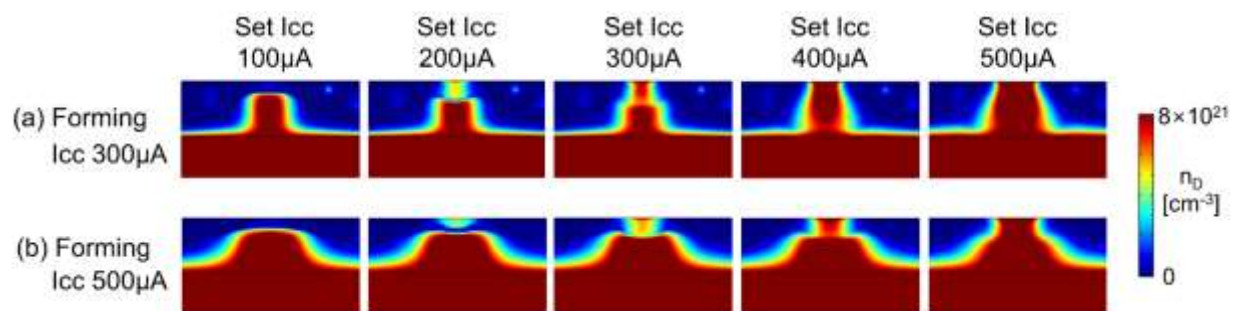


Figure S6. 2-D maps of n_D profile after Set, for different Set I_{CC} s. The devices first goes through (a) Forming with 300 μA I_{CC} and 1st Reset, or (b) Forming with 500 μA I_{CC} and 1st Reset, respectively. Different V_O configurations are observed during Set, for devices formed at these different Forming conditions, even if the same Set I_{CC} s are used.

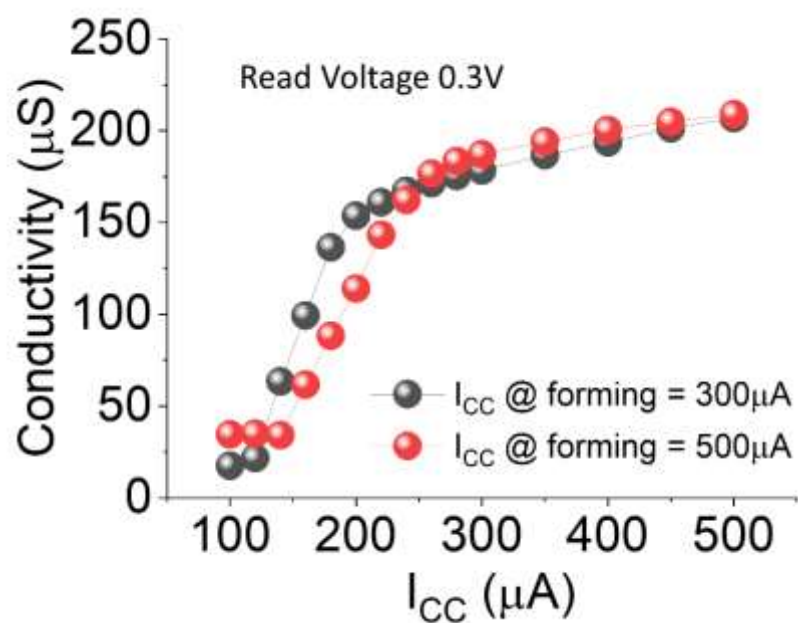


Figure S7. Simulated LTP behaviors with incremental I_{CC} , for different Forming conditions. Forming with lower I_{CC} provides a larger dynamic range, but a steeper slope which may make it more difficult to fine-tune the conductance.

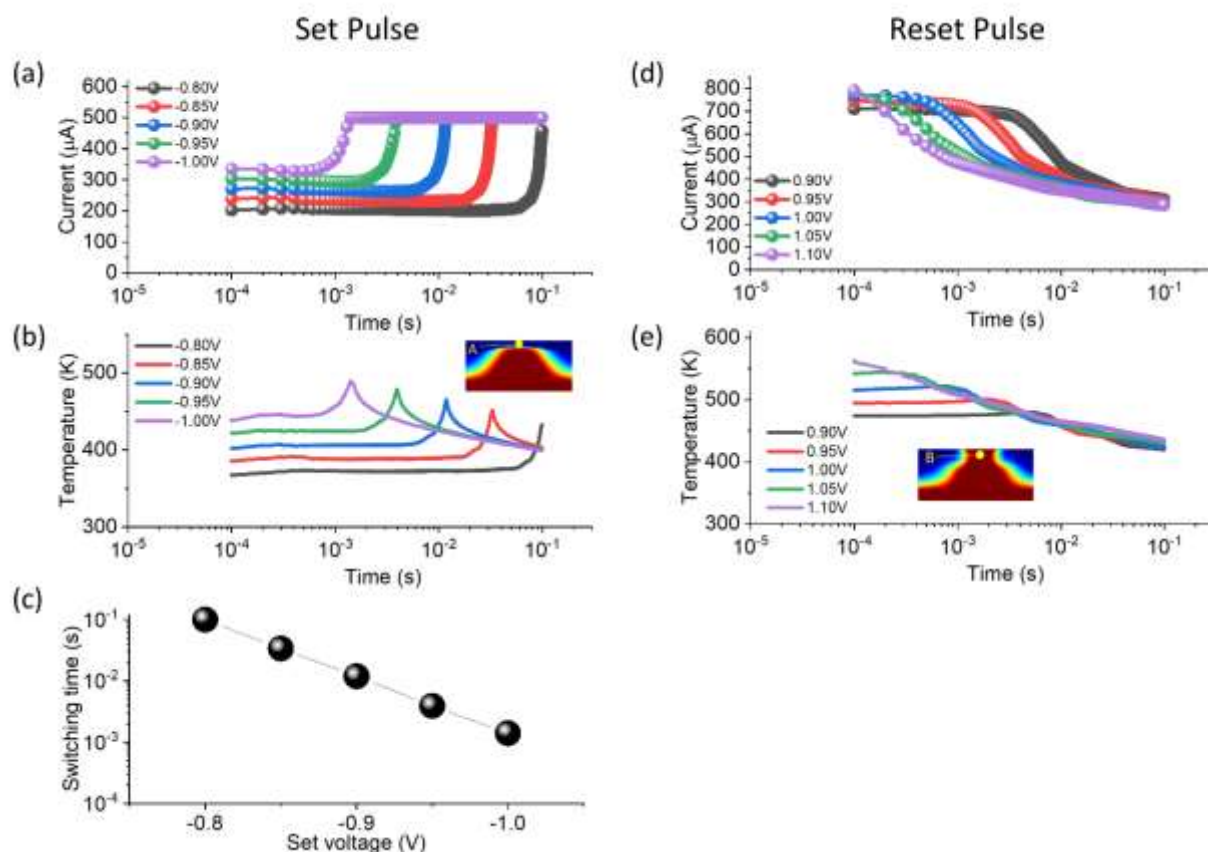


Figure S8. Set and Reset pulse simulation. (a) Current and (b) temperature evolutions at the location 'A' at different Set voltages. (c) Set speed vs. Set voltage. (d) Current and (e) temperature evolutions at the location 'B' at different Reset voltages.

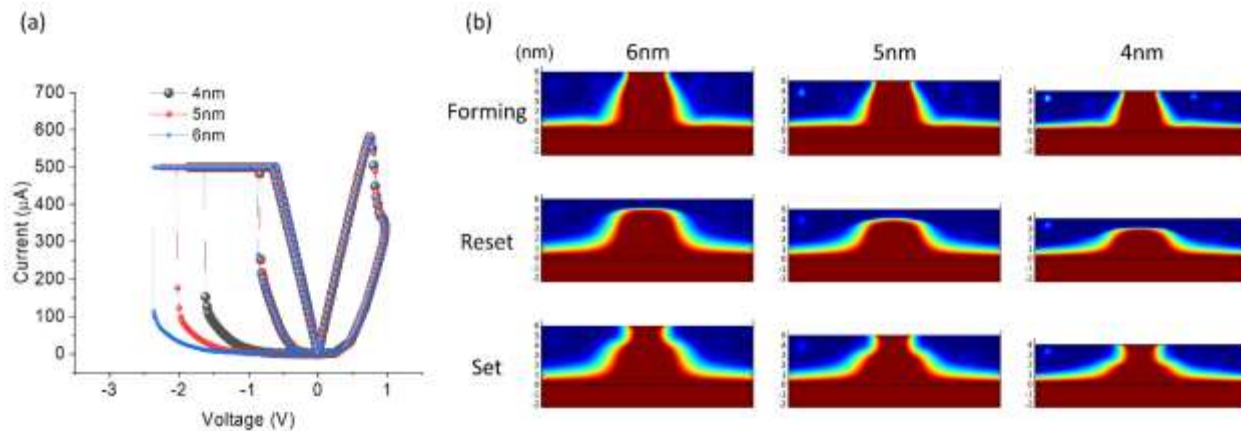


Figure S9. Forming and switching characteristics vs. Ta₂O₅ film thickness. (a) I-V characteristics showing the Forming, and subsequent Reset and Set processes. (b) 2-D maps of n_D profile after Forming, first Reset, and subsequent Set, for devices with different Ta₂O₅ film thicknesses.

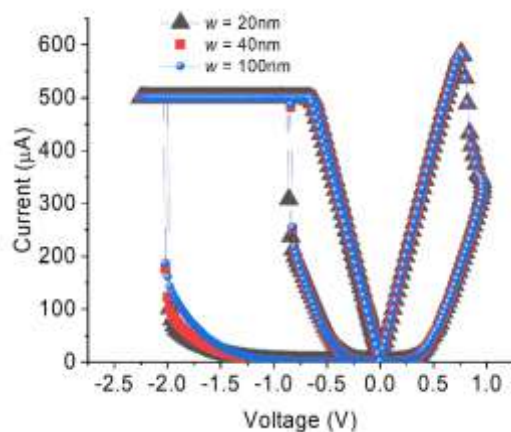


Figure S10. Forming and switching characteristics vs. device area (20nm, 40nm, and 100nm).

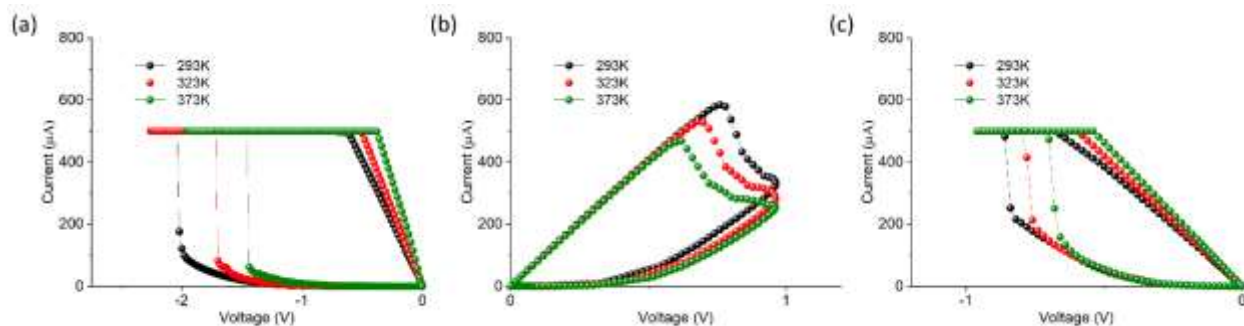


Figure S11. Forming/Reset/Set switching characteristics as a function of ambient temperature, showing the (a) Forming, (b) first Reset, and (c) subsequent Set processes at different ambient temperatures.

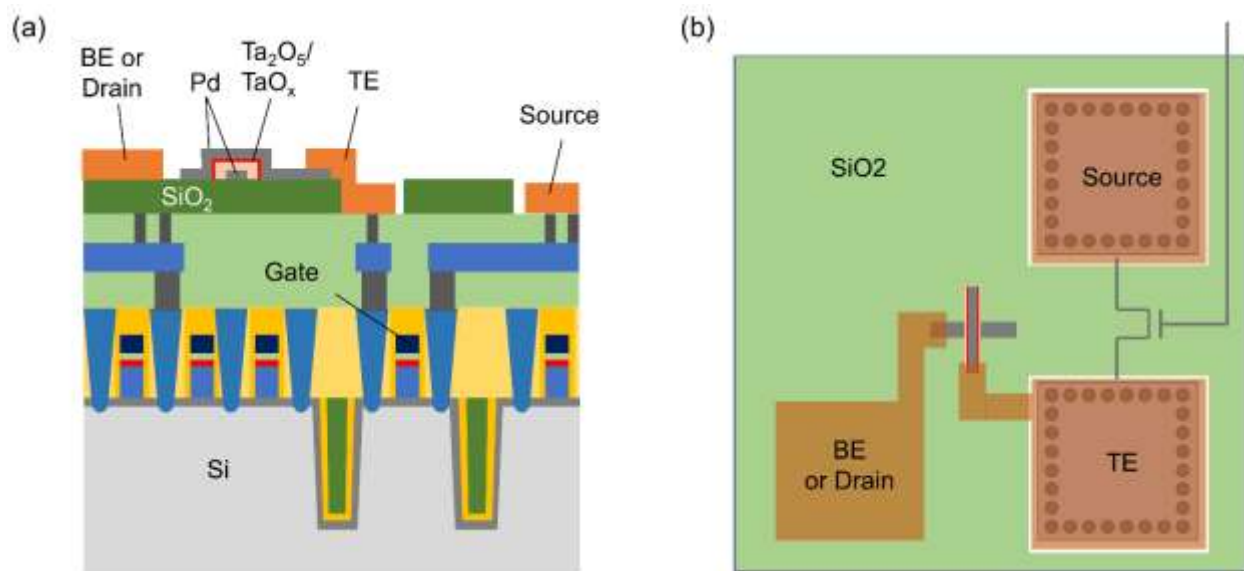


Figure S12. Schematic of the 1T1R structure. (a) Cross-sectional view of the 1T1R structure with Pd/Ta₂O₅/TaO_x/Pd bilayer memristor device. (b) Top view. The BE of the memristor is connected to the Drain of the transistor to provide negative Forming and Set voltages to the memristor device.

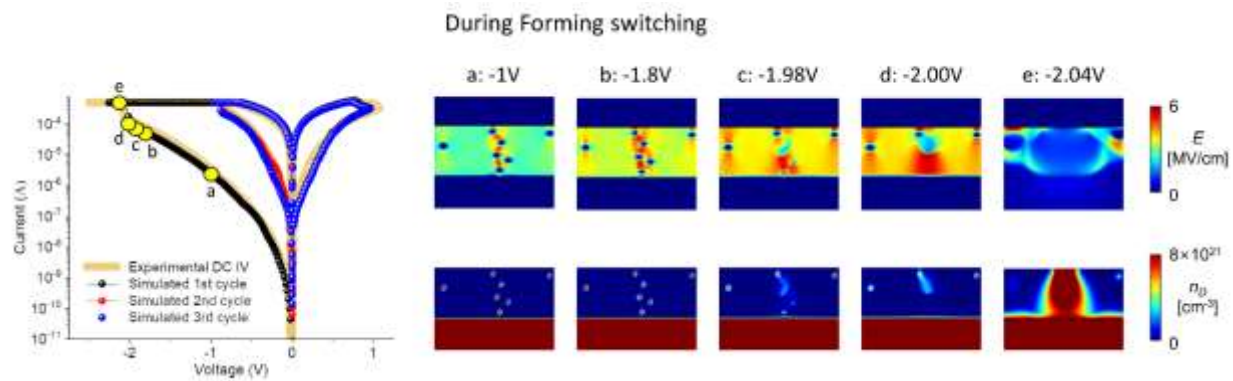


Figure S13. Evolution of internal electric field distribution, at different stages of the Forming process.

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